QualityForge 全项目产品需求文档

AI PRD

文档类型	版本	状态	日期
PRD	v1.0		2026-08-10

文档定位
QualityForge AI PRD OpenAPI DB AI

0.

	QualityForge AI
	/
	AI
	ISO/IEC/IEEE 29148 PRD Given/When/Then
	OpenAPI

0.1

- 1
- 2
- 3
- 4
- 5
- 6
- 7 AI Agent Skill
- 8
- 9
- 10

- 11
- 12
-

1.

1.1

QualityForge AI AI CI/CD
AI

1.2

-
-
- Locator
- Trace
- CI
- AI oracle

1.3

	RequirementDocument → TestingIR → BusinessRule → Scenario → DesignTestCase → Assertion → Coverage
	AI Secret fail-closed
	Agent
	API DB

2.

“ ” Runner PRD

		AI

/		/ DesignTestCase ReviewTask Coverage
	Runner	Secret
		AI
/	Runner Secret	
AI/Skill	Agent Prompt Skill	

3.

3.1

Skill Agent CI/CD	CI
API Web UI	
Jira/Git/CI/	
AI	

3.2

		80%	
	/P95	80%	oracle
	Top-K	95%	Recall/Precision
CI	TIA	60%	
	sourceTrace		Gate 0 100%
AI	AI		Skill/ /

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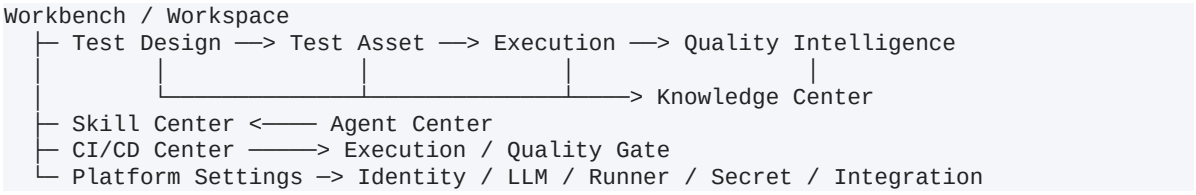
oracle 80%/95%/60%

4.



Workbench	AI
Workspace	Secret OpenAPI
Test Design	IR/
Test Asset	Locator Mock
Execution	Runner /
Quality	Coverage
AI	Skill Agent Prompt Workflow Trace
Knowledge / CI / Settings	

5.

5.1

- Workspace
- RequirementDocument
- REJECT PASS_WITH_REVIEW ReviewTask PASS
- TestingIR BusinessRule SourceTrace
- TestDesignPlan TestScenario DesignTestCase Assertion
- Coverage TestCase

5.2

- TestCase TestSuite ExecutionPlan Runner
- REST WebSocket
- ExecutionResult Trace
- Quality Intelligence Coverage RCA

- Knowledge Center CI/CD QualityGate

5.3 Agent/Skill Gate 0

```
Supervisor → Requirement Quality Gate → / /
→ Requirement Compiler → Independent IR Validator → / / /
→ Assertion Verifier → Coverage Analyzer → HITL Gatekeeper →
```

6.

6.1 Workspace

AI

V1		Workspace WorkspaceEnvironment WorkspaceSecret WorkspaceAIContext	workspaceId/tenantId

ID			
QF-FR-WS-001	P0		
QF-FR-WS-002	P0		
QF-FR-WS-003	P0		Dev/Test/Staging Secret
QF-FR-WS-004	P1	AI	OpenAPI/Swagger

6.2 Test Design

Testing IR

V1		RequirementDocument TestingIR BusinessRule TestDesignPlan TestScenario DesignTestCase Assertion ReviewTask	TestCase REJECT/BLOCK_HUMAN

ID			
QF-FR-TD-001	P0		PDF Word Markdown OpenAPI
QF-FR-TD-002	P0		NFR
QF-FR-TD-003	P0	Testing IR	IR Source Trace
QF-FR-TD-004	P0		
QF-FR-TD-005	P0		DesignTestCase Assertion TestCase
QF-FR-TD-006	P1		

6.3 Test Asset

Mock

V1		TestCase TestSuite Assertion Locator TestDataProfile APIContract MockConfig	AI

ID			
QF-FR-TA-001	P0		TestCase CRUD CSV
QF-FR-TA-002	P0		
QF-FR-TA-003	P1	Locator	
QF-FR-TA-004	P1		
QF-FR-TA-005	P1	Mock	API Contract /

6.4 Execution

API/UI Trace

V1	Runner	ExecutionPlan ExecutionTask ExecutionResult RunnerNode	HTTP

ID			
QF-FR-EX-001	P0		CI
QF-FR-EX-002	P0	Runner	Runner
QF-FR-EX-003	P0		API/UI
QF-FR-EX-004	P0		WebSocket
QF-FR-EX-005	P1		

6.5 Quality Intelligence

V1 / V2		QualityAnalysisReport FailureAnalysisDetail FailureCluster CoverageReport	AI

ID			
QF-FR-QI-001	P0		
QF-FR-QI-002	P0		Trace
QF-FR-QI-003	P0		CoverageReport
QF-FR-QI-004	P1	RCA	
QF-FR-QI-005	P2		

6.6 Skill Center

Playwright HTTP Skill

V1 / V2	Skill	Skill SkillInvocation	/ / PASS allowlist Skill

ID			
QF-FR-SK-001	P0	Skill	API Testing Web Playwright
QF-FR-SK-002	P0		Skill Must Not
QF-FR-SK-003	P1		Skill eval
QF-FR-SK-004	P1		Secret CPU

6.7 Agent Center Agent

Agent Prompt Trace

Gate 0 / V2	AI	Agent AgentPromptVersion AgentSession AgentTraceNode	Agent

ID			
QF-FR-AG-001	P0	Agent	Prompt Skill allowlist
QF-FR-AG-002	P0	Trace	
QF-FR-AG-003	P0	Gate 0	1 Supervisor + 3 Subagent + 13 Skill DAG
QF-FR-AG-004	P1		DAG

6.8 Knowledge Center

RAG /UI/ Agent

V1		KnowledgeItem KnowledgeVector BaselineItem	

ID			
QF-FR-KN-001	P0		
QF-FR-KN-002	P0		AI
QF-FR-KN-003	P0		API Schema
QF-FR-KN-004	P1		Testing Pattern

6.9 CI/CD Center

TIA

V1 / V2		PipelineConfig PipelineRun QualityGate	AI

ID			
QF-FR-CI-001	P0	Webhook	Jenkins/GitLab/GitHub
QF-FR-CI-002	P0		Passed/Warned/Blocked
QF-FR-CI-003	P1	TIA	Git Diff
QF-FR-CI-004	P1		

6.10 Platform Settings

LLM Runner Secret

V1	Runner	User Role LlmConfig RunnerNode WorkspaceSecret IntegrationConfig	

ID			
QF-FR-PS-001	P0		RBAC

ID			
QF-FR-PS-002	P0	LLM	
QF-FR-PS-003	P0	Runner	
QF-FR-PS-004	P0	Secret	

7. AI Agent Skill

7.1

L0	AI /	
L1	AI	
L2		allowlist
L3		Gate 0

7.2 Agent

Agent		Skill	
qf-supervisor		plan-orchestrator	
qf-requirement-ir	IR		IR
qf-test-design			IR
qf-independent-verifier		IR Coverage HITL	

7.3

- docs Skill Golden
- deny_all Secret
- Skill Schema allowlist
- 1 Schema fail-closed
- Prompt
-

8.

8.1

- tenant/workspace
- / SourceTrace
- DesignTestCase TestCase

- CoverageReport
- /Prompt/Skill

8.2

REST	/api/v1 camelCase JSON Bearer/API Key/Runner Token
WebSocket	Execution AI Agent Trace Workspace seq
CI/CD	Webhook
	URL
LLM	

9.

	V1	/
	50+ API FCP < 3.0s AI P95 < 20s	Runner
	AES-256-GCM RBAC	RLS/ Secret
	1 Runner	V2
	JSON Metrics	requestId/runId/traceId Agent
		Saga
		WCAG 2.2 AA +
		Prompt

10.

Gate 0 / V0		→IR→ → → → →Coverage→	
V1 1–3		Workspace / API/UI Skill/Agent RAG/Webhook	Runner Skill Agent
V2 4–6		Locator RCA Skill Workflow TIA/	
V3 7–12		Runner Skill Agent PR	

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B-1/B-2/B-3	2026-07-31	2026-08-04	B-4/B-5/B-6	Sprint 0
OpenAPI operationId AI /				

11.

11.1

AC-E2E-01

Given **When** Gate IR **Then** Gate PASS TestCase

AC-SEC-01

Given A B **When** A Token B ID **Then** 403 404

AC-AI-01

Given Agent IR/ / **When** Agent **Then**

AC-EX-01

Given Runner **When** Runner **Then** HTTP 2xx PASS

AC-CI-01

Given **When** **Then**

11.2

- ID /API/ / 2xx
- OpenAPI operationId / / / / /
- PRD API DB
- AI oracle Golden Set Prompt Skill
- NFR
- /

12.

ID			
DEC-01			RBAC /API/DB
DEC-02	164 OpenAPI operationId	SDK	Sprint 0 operationId
DEC-03	AI	PASS/REVIEW	Golden Set
DEC-04	LLM oracle		checker + +

ID			
DEC-05		ROI	
DEC-06			Gate 0/V1 V2/V3
DEC-07			

A.

	54- .md	
	03- Glossary.md	AI
	04- .md	
	10–19 PRD	
AI	21-AI .md	L0–L3
Agent/Skill	55-Agent Skill SOP.md	allowlist
API	32-API .yaml	REST Schema
DB	33- Schema .md	
/	34- .md / 36-WebSocket .md	RBAC/RLS/Secret
	docs/05-audits/current/	

03/04 PRD 04 32 33 AI 21 03 04 32 33 PRD